

Hands-on Lab Evaluation – 2 (Module II)

Topic

Regression Models

Objective

To understand, implement, and compare different regression algorithms for predicting continuous values using Machine Learning. Students will evaluate model performance using regression metrics and identify the most suitable model for real-world prediction problems.

Dataset

House Price Prediction Dataset

Dataset Links

- Kaggle: <https://www.kaggle.com/datasets/harishkumardatalab/housing-price-prediction>
- Alternative Dataset (House Prices Competition):
<https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques>

Course Outcomes (COs) Achieved

- CO3: Implement various supervised Machine Learning models.
- CO4: Train and test Machine Learning models using real-world datasets.

Bloom's Level Achieved

- BL3 – Apply
- BL4 – Analyze

Problem Statement

A real estate company wants to develop a Machine Learning model to estimate the selling price of houses based on various property features such as area, number of bedrooms, bathrooms, furnishing status, parking spaces, and location. Students are required to preprocess the dataset, implement different regression algorithms, compare their performance, and identify the most suitable model for accurate house price prediction.

Tasks to be Performed

1. Load the House Price Prediction Dataset using Pandas. Display the first and last 10 records, dataset information, summary statistics, identify missing values, and visualize the distribution of numerical features using Histograms and Boxplots.
2. Perform data preprocessing by handling missing values, encoding categorical variables using Label Encoding or One-Hot Encoding, applying feature scaling using StandardScaler or MinMaxScaler, and splitting the dataset into Training (70%) and Testing (30%) datasets.
3. Implement Simple Linear Regression and Multiple Linear Regression models. Compare their prediction performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.
4. Implement Polynomial Regression, Ridge Regression, and Lasso Regression. Compare their prediction accuracy and discuss the effect of regularization on model performance.

5. Implement Support Vector Regression (SVR) and compare its performance with all previously implemented regression models using MAE, MSE, RMSE, and R^2 Score. Summarize your observations and recommend the best-performing regression model for house price prediction.

Submission Deliverables

- Jupyter Notebook (.ipynb)
- Python Source Code (.py)
- Cleaned Dataset (.csv)
- Regression Model Comparison Report
- Data Visualization Graphs
- Prediction Graphs
- Output Screenshots
- Short Conclusion (150–200 words)

Evaluation Rubrics (Total: 5 Marks)

Evaluation Criteria	Marks
Data Preprocessing & Regression Model Implementation	2
Model Comparison using Regression Metrics	1
Visualization, Code Quality & Documentation	1
Viva-Voce	1
Total	5

Expected Learning Outcomes

- Understand the concept of regression in Machine Learning.
- Preprocess real-world datasets for regression analysis.
- Implement and compare multiple regression algorithms.
- Evaluate regression models using standard performance metrics.
- Select the most appropriate regression model based on experimental results.
- Interpret prediction outcomes and visualize regression performance effectively.

Software & Libraries Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn